Harshitha Menon

Computer Scientist, Center for Applied Scientific Computing
Lawrence Livermore National Laboratory

☎ *650.741.4260*
✉ *harshitha@llnl.gov*
http://harshithamenon.com
Google Scholar page

Education

2016 **Ph.D., Computer Science**, *University of Illinois at Urbana-Champaign*.
Adaptive Load Balancing for HPC Applications. Advisor: Laxmikant V. Kale

2012 **M.S., Computer Science and Engineering**, *University of Illinois at Urbana-Champaign*.

2006 **B.Tech., Computer Science and Engineering**, *College of Engineering*, Trivandrum, India.

Research and Professional Experience

Oct 2018–Present **Lawrence Livermore National Laboratory**, *Computer Scientist*.
Center for Applied Scientific Computing (CASC)

- **Lead PI, Ellora Project (DOE ASCR AI for Science):** Leading a $7M multi-institutional project on AI-assisted HPC software development with 20+ researchers across 4 institutions (ORNL, Univ. Maryland, Northeastern Univ.). Developing large language model approaches for code generation, performance prediction, multi-modal LLMs, performance-aligned LLMs, and interpretability techniques on HPC code. Responsibilities include technical vision, cross-institution collaboration, budget authority ($950K/year LLNL, $2.35M/year total), developing tools and techniques, obtaining computational resources (ALCC allocation), and reporting to Program Managers.
- **MADA Project (DOE ASC):** Main contributor for the Multi-Agent Design Assistant framework, providing technical direction for multi-agent framework and LLM integration with physics simulation, meshing tools, surrogate models, and HPC systems via Flux.
- **CADSAT Project (DOE AI4NS):** Co-PI for CAD to Simulation Agent Toolkit, leveraging AI-enabled agents to automate design-to-analysis engineering workflows.
- **Spack Package Manager (BUILD LDRD-SI Thrust Lead):** Led the machine learning thrust to optimize automatic software package integration. Developed Graph Neural Network models to predict package build outcomes and applied masked-based pre-training for limited data scenarios. Created probabilistic approaches for selecting build configurations.
- **Approximate Computing (PI, LLNL LDRD, 2019–2022):** Led research on mixed-precision analysis using automatic differentiation (ADAPT), uncertainty quantification for approximate computing, framework for approximate computing.
- **ZFP Compression Research:** Lead developer of the ZFPoly library, which reduces storage and improves accuracy by six orders of magnitude compared to IEEE floating point.
- **Computing Postdoc Advocate and Institutional Postdoc Program Board:** Responsible for maintaining an effective postdoc program within Computing and the lab.

Oct 2016–Oct 2018 **Lawrence Livermore National Laboratory**, *Postdoctoral Research Staff*.
Center for Applied Scientific Computing (CASC)

- Developed techniques for error analysis in HPC applications using automatic differentiation.
- Mentored graduate students and summer interns.

Aug 2012–Aug 2016 **Department of Computer Science, University of Illinois at Urbana-Champaign**, *Research Assistant*.
Parallel Programming Laboratory, Advisor: Prof. Laxmikant V. Kale

- Developed distributed load balancing strategies for HPC applications; awarded Best Student Paper Finalist at SC'13.
- Created measurement-based and topology-aware load balancing algorithms for irregular applications.
- Contributed to the Charm++ parallel programming framework.
- Improved performance of ChaNGa cosmology simulation at 512K cores on Blue Waters, pushing the scalability limits of N-body simulations; work published in Computational Astrophysics and Cosmology and in Monthly Notices of the Royal Astronomical Society.

Summer 2015 **Charmworks**, *Advanced Software Developer Intern*.

- Developed features for Charm++ runtime system.

Summer 2013 **Lawrence Livermore National Laboratory**, *Research Intern*.
Developed graph partitioning-based load balancing algorithms. Received Best Poster Award at LLNL Student Poster Symposium.

Aug 2010–Dec 2011 **Department of Computer Science, University of Illinois at Urbana-Champaign**, *Teaching Assistant*.
- Teaching Assistant for CS 225: Data Structures.
- Ranked as "Teachers Ranked as Excellent" by students (Fall 2011).

Summer 2011 **Google**, *Summer Intern*.
- Worked on distributed data processing systems called MapReduce.

Jul 2006–Jul 2010 **Google**, *Software Engineer*.
- Worked on Gmail and MapReduce systems.

Grants and Funded Projects

External Funding

2026–Present **LLNL PI**, AI4HPC: An Iterative Framework for SI-Assisted Scientific Software Development and Optimization. DOE Genesis Mission, Phase II. Led by Argonne National Laboratory (ANL).

2025–Present **Co-PI**, AI4HPC Genesis ModCon Model Seed Team. DOE Office of Science Program. PI: Andrew Siegel (ANL).

2025–Present **Co-PI**, CADSAT: CAD to Simulation Agent Toolkit. DOE AI4NS Program. PI: Steven Owen (Sandia National Laboratory). Project aimed at leveraging AI-enabled agents to automate design-to-analysis engineering workflows.

2024–2027 **Lead PI**, Ellora: Productive AI-Assisted HPC Software Ecosystem. Total: $2,350,000 (LLNL: $950K/year). DOE Office of Science ASCR AI for Science program. Co-PIs: William Godoy (ORNL), Abhinav Bhatele (Univ. Maryland), Tom Goldstein (Univ. Maryland), Arjun Guha (Northeastern Univ.), David Bau (Northeastern Univ.).

2020–2024 **Co-Investigator**, d-FEM: Differentiable Finite Element Methods. DOE ASCR. PI: Tzanio Kolev.

Internal Funding

2025–Present **Co-Investigator**, C4ML: Compression for Machine Learning. Lawrence Livermore National Laboratory LDRD. PI: Peter Lindstrom.

2024–Present **Investigator**, MADA: Multi-Agent Design Assistant. NNSA ASC Program.

2021–2024 **Thrust Lead**, BUILD: Binary Understanding and Integration Logic for Dependencies. Lawrence Livermore National Laboratory LDRD-SI. PI: Todd Gamblin. Led the thrust on machine learning to predict high-quality build configurations.

2019–2022 **PI**, ApproxHPC: Approximate Computing for Energy-Efficient HPC. Lawrence Livermore National Laboratory LDRD-ER. Budget: $690K/year.

2018 **PI**, LEARN: Fault Tolerance for Exascale Computing. DOE ATDM in collaboration with LANL. $112K.

Computing Time Allocations

2025 **PI**, ALCC Award: Training Multi-modal Models for HPC Code and Data. 700,000 Node-Hours on Frontier, Aurora, and Perlmutter.

2024 **Co-PI**, INCITE Award: Scaling Distributed AI Training and Developing New AI Models. 750,000 Node-Hours on Frontier, Polaris, and Aurora. PI: Abhinav Bhatele (Univ. Maryland).

Awards and Honors

2026 **Director's Early and Mid-Career Recognition Program Award** (Early Career category), Lawrence Livermore National Laboratory

2026 Featured in SC Conference Women's History Month Initiative, Supercomputing (SC)'26

2025 **Best Poster Award Finalist**, Supercomputing (SC)'25. Md Mahbubur Rahman, Arjun Guha, and Harshitha Menon. "Understanding LLM Behavior on HPC Data via Mechanistic Interpretability."

2022 DDS&T Excellence in Publication Award for "Approximate Computing Through the Lens of Uncertainty Quantification," SC'22 (ApproxHPC Project; mentees Konstantinos Parasyris and James Diffenderfer won in Postdoc Category)

2021 **Best Reproducibility Advancement Award**, Supercomputing (SC)'21

2019 Outstanding Mentorship Award, Lawrence Livermore National Laboratory

2019 Women's Early Career Accelerator, NVIDIA GTC Conference

2017 Best Poster Award Finalist, Supercomputing (SC)'17. Vivek Kale, Harshitha Menon, and Karthik Senthil. "Adaptive Loop Scheduling with Charm++ to Improve Performance of Scientific Applications."

2016 Featured Article, IEEE Computer Magazine (October 2016)

2014 **ACM/IEEE-CS George Michael Memorial HPC Fellowship**, SC'14. One of two recipients worldwide for research on "Scalable Load Balancing and Adaptive Run Time Techniques."

2014 **Google Anita Borg Memorial Scholarship**

2014 Best Paper Award, IEEE Cluster'14

2014 Feng Chen Memorial Best Paper Award, University of Illinois at Urbana-Champaign

2013 **Best Student Paper Award Finalist**, Supercomputing (SC)'13. Harshitha Menon and Laxmikant Kalé. "A Distributed Dynamic Load Balancer for Iterative Applications."

2012 **Siebel Scholarship** ($35,000), University of Illinois at Urbana-Champaign. Recognizes top graduate students from leading graduate schools worldwide in business, computer science, bioengineering, and energy science.

2012 HPC Challenge Class II Award Finalist, Supercomputing (SC)'12

2011 Teachers Ranked as Excellent, University of Illinois at Urbana-Champaign

2010 **Google Fellowship for Employees**

2007 Google Founders Award, part of Gmail team

Tutorials and Teaching

2019 **PEARC**, *Tutorial on Floating-Point Analysis Tools*,
Ignacio Laguna, Michael Bentley, Ian Briggs, Ganesh Gopalakrishnan, Harshitha Menon, and Cindy Rubio Gonzalez.

2019 **SC**, *Floating-Point Analysis and Reproducibility Tools for Scientific Software*,
Ignacio Laguna, Michael Bentley, Ian Briggs, Ganesh Gopalakrishnan, Hui Guo, Michael O. Lams, Harshitha Menon, Pavel Panchekha, and Cindy Rubio Gonzalez.

2010 - 2011 **UIUC**, *CS 225 - Data Structures*,
Teaching Assistant.

Mentoring and Advising

Postdoctoral Researchers

2025–Present **Daniel Nichols**, Sidney Fernbach Fellow, Lawrence Livermore National Laboratory.

2025–Present **Gautam Singh**, Lawrence Livermore National Laboratory.

2020–2022 **Konstantinos Parasyris**, Lawrence Livermore National Laboratory.

2020–2022 **James Diffenderfer**, Lawrence Livermore National Laboratory.

Graduate Students (Co-advised/Mentored)

2022–2024 **Daniel Nichols**, University of Maryland. Advisor: Prof. Abhinav Bhatele. Now: Fernbach Fellow at LLNL.

2022–2024 **Caetano Melone**, Stanford University. Now: Staff at LLNL.

2021 **Manisha Mukherjee**, Carnegie Mellon University.

2016–2023 **Zhimin Li**, University of Utah.

2019–2022 **Jackson Vanover**, UC Davis.

2019 **Diego Jimenez Vargas**, CENAT, Costa Rica.

2018 **Nathan Pinnow**, Western Washington University.

Undergraduate Students

2023 **Alec Scott**, University of Arizona. *LLNL Intern.*

2022 **Garima Singh**.

2021 **Baidyanath Kundu**.

2020 **Logan Moody**, James Madison University. *LLNL Intern.*

2019 **Garrett Folks**, James Madison University. *LLNL Intern.*

Professional Service

Conference and Workshop Organization

2025 **Area Chair (AI Track)**, Supercomputing India (SC-India) 2025

2025 **Organizer**, Workshop on LLMs for HPC (LLM4HPC), PPoPP 2025

2025 **Organizer**, Mini-symposium on Machine Learning for Scientific Workflows (ML4SW), PASC 2025

2024 **Organizer**, Workshop on Efficient Systems for Foundation Models and LLMs (ESwML), EuroSys 2024

2024 **Organizer**, Mini-symposium on Machine Learning for Scientific Workflows (ML4SW), PASC 2024

2023 **Organizer**, Workshop on High-Level Parallel Programming Models and Supportive Environments (HIPS), IPDPS 2023

2022 **Workshop Chair**, IEEE Cluster 2022, Heidelberg, Germany

Technical Program Committees

2026 SC'26, HPDC'26

2025 SC'25, HPDC'25, PPoPP'25, IPDPS'25, ICPP'25

2024 SC'24, PPoPP'24, HPDC'24, ISC'24

2023 PPoPP'23, HPDC'23, HiPC'23

2022 HPDC'22, HiPC'22, IPDPS'22

2021 ICPP'21, HiPC'21

2020 Cluster'20, ICPP'20

2019 CARLA'19, PASC'19, Euro-Par'19, PPoPP'19

2018 ICPP'18, Euro MPI'18

Editorial and Review Boards

2020–2021 **Technical Review Board Member**, IEEE Transactions on Parallel and Distributed Systems (TPDS)

Institutional Service

2026 **Technical Lead**, Computing External Review Committee, Lawrence Livermore National Laboratory

2022–Present **Board Member**, Institutional Postdoc Program Board, Lawrence Livermore National Laboratory

2023–2025 **LDRD ER Review Committee**, Lawrence Livermore National Laboratory

Other Conference Committees

2021 **Workshop Committee**, Supercomputing (SC)'21

2020 **Research Posters Committee**, Supercomputing (SC)'20

Journal Reviewing

IEEE Transactions on Parallel and Distributed Systems (TPDS), ACM Transactions on Architecture and Code Optimization (TACO), International Journal of High Performance Computing Applications (IJHPCA), Concurrency and Computation: Practice and Experience, Performance Modeling, Benchmarking and Simulation (PMBS), Future Generation Computer Systems (FGCS)

Panels

Nov 2022 **Panelist**, "Approximate Computing", Supercomputing (SC)'22

Nov 2022 **Panelist**, "Careers at HPC", Supercomputing (SC)'22

Aug 2022 **Panelist**, "Reproducibility in HPC", UC Santa Cruz Open Source Symposium 2022

Software Projects

MADA Multi-Agent Design Assistant framework for AI-driven scientific workflows.

ZFPoly Compressed-vector library for floating-point data with polynomial basis.

ADAPT Algorithmic Differentiation based method for mixed precision analysis.

Software Contributed to

HPAC Approximate Computing framework for HPC applications.

Charm++ Object-based message driven parallel programming paradigm.

ChaNGa Cosmology simulation application.

MapReduce Distributed data processing.

Invited Talks, Panels, and Tutorials

Invited Talks

2025 **Toward Next-Generation Ecosystems for Scientific Computing Workshop**, *AI-Assisted HPC Software Development*.
Organized by Lois McInnes (DOE Office of Science Distinguished Scientist Fellow)

2025 **DOE Computer Science PI Meeting**, *Ellora: AI-Assisted HPC Software Ecosystem*.

Nov 2024 **SC'24 BSSW Workshop**, *AI for Scientific Software Development*.

Nov 2024 **AI for Science Workshop, ORNL**, *LLMs for HPC Code Development*.

Feb 2024 **Workshop on Differentiable Parallel Programming, PPoPP'24**, *Automatic Differentiation for Correctness and Correctness of Automatic Differentiation*.

Mar 2024 **Minisymposium on Reproducibility, SIAM-PP'24**, *Ensuring Reproducibility Amidst Approximate Computing*.

Jun 2023 **DOE/NSF Workshop on Correctness in Scientific Computing 2023**, *Ensuring Correctness in Programs Generated by LLM: Challenges and Solutions*.

Dec 2021 **Differentiable Programming Workshop, NeurIPS'21**, *Approximate High Performance Computing Guided by Automatic Differentiation*, Virtual.

Mar 2021 **New Approaches and Challenges for Auto-Tuning in the Exascale Era, SIAM-PP'21**, *Auto-Tuning Parameter Choices in HPC Applications using Bayesian Optimization*, Virtual.

Aug 2020 **23rd European Workshop on Automatic Differentiation (EuroAD)**, *Plenary Talk: Error Analysis Using Automatic Differentiation in HPC Applications*, Virtual.

May 2020 **ICERM Workshop on Variable Precision in Mathematical and Scientific Computing**, *Automatic Mixed Precision Analysis and Transformation Tool*, Brown University (Virtual).

2021 **Argonne National Laboratory**, *Approximate HPC: A Fast and Energy Efficient Computing Paradigm in the Post-Moore Era*.

2019 **James Madison University**, *Adaptive Techniques for High-Performance Computing*.
Invited departmental seminar

Oct 2023 **PackagingCon 2023**, *Using Graph Neural Networks to Predict Build Success in Software Ecosystems*, Berlin, Germany.

Panels

Nov 2025 **Supercomputing (SC)'25**, *Panelist: Research Software Engineering in the Age of AI*, St. Louis, MO.
With Daniel S. Katz, Alfredo Goldman, Michael Heroux, Rohan Marwaha, Shailaja K. Sampat

Nov 2022 **Supercomputing (SC)'22**, *Panelist: Approximate Computing*.

Nov 2022 **Supercomputing (SC)'22**, *Panelist: Careers at HPC*.

Aug 2022 **UC Santa Cruz Open Source Symposium 2022**, *Panelist: Reproducibility in HPC*.

Tutorials

Nov 2019 **Supercomputing (SC)'19**, *Floating-Point Analysis Tools Tutorial*, Denver, CO.
Co-presenters: Ignacio Laguna, Michael Lam

Birds of a Feather (BoFs)

Nov 2024 **Supercomputing (SC)'24**, *AI for HPC Software Development*, Atlanta, GA.

Nov 2024 **Supercomputing (SC)'24**, *Approximate Computing for HPC*, Atlanta, GA.

Publications

Mentored graduate co-authors are indicated with ($^+$) in the author list; mentored post-doctoral researchers are indicated with (*) in the author list.

Preprints and Papers Under Review

[1] Gregory Bolet, Giorgis Georgakoudis, Konstantinos Parasyris, **Harshitha Menon**, Niranjan Hasabnis, Kirk W Cameron, and Gal Oren. Counting without running: Evaluating llms' reasoning about code complexity. *arXiv preprint arXiv:2512.04355*, 2025.

[2] Caetano Melone, Daniel Nichols*, Konstantinos Parasyris, Todd Gamblin, and **Harshitha Menon**. Llms as packagers of hpc software. *arXiv preprint arXiv:2511.05626*, 2025.

[3] **Harshitha Menon**, Charles F Jekel, Kevin Korner, Brian Gunnarson, Nathan K Brown, Michael Stees, M Giselle Fernandez-Godino, Walter Nissen, Meir H Shachar, Dane M Sterbentz, et al. Multi-agent collaboration for automated design exploration on high performance computing systems. *arXiv preprint arXiv:2603.11515*, 2025.

[4] Daniel Nichols*, Konstantinos Parasyris, Charles Jekel, Abhinav Bhatele, and **Harshitha Menon**. Integrating performance tools in model reasoning for gpu kernel optimization. *arXiv preprint arXiv:2510.17158*, 2025.

[5] Daniel Nichols*, Konstantinos Parasyris, **Harshitha Menon**, Brian R Bartoldson, Giorgis Georgakoudis, Tal Ben-Nun, and Abhinav Bhatele. Modeling code: Is text all you need? *arXiv preprint arXiv:2507.11467*, 2025.

[6] Daniel Nichols*, Prajwal Singhania, Charles Jekel, Abhinav Bhatele, and **Harshitha Menon**. Optimizing agentic language model inference via speculative tool calls. *arXiv preprint arXiv:2512.15834*, 2025.

[7] Meir H Shachar†, Dane M Sterbentz†, Harshitha Menon†, Charles F Jekel†, M Giselle Fernández-Godino, Nathan K Brown, Ismael D Boureima, Yue Hao, Kevin Korner, Robert Rieben, et al. Multi-agent design assistant for the simulation of inertial fusion energy. *arXiv preprint arXiv:2510.17830*, 2025. †Equal contribution.

[8] Prajwal Singhania, Siddharth Singh, Lannie Dalton Hough, Akarsh Srivastava, **Harshitha Menon**, Charles Fredrick Jekel, and Abhinav Bhatele. Llm inference beyond a single node: From bottlenecks to mitigations with fast all-reduce communication. *arXiv preprint arXiv:2511.09557*, 2025.

Peer Reviewed Conference & Journal Papers

[9] Joshua Hoke Davis, Pranav Sivaraman, Joy Kitson, Konstantinos Parasyris, **Harshitha Menon**, Isaac Minn, Giorgis Georgakoudis, and Abhinav Bhatele. Taking gpu programming models to task for performance portability. In *Proceedings of the 39th ACM International Conference on Supercomputing (ICS)*, pages 776–791, 2025.

[10] Daniel Nichols*, Pranav Polasam, **Harshitha Menon**, Aniruddha Marathe, Todd Gamblin, and Abhinav Bhatele. Performance-aligned llms for generating fast code. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 2025.

[11] Yangtian Zi, **Harshitha Menon**, and Arjun Guha. More than a score: Probing the impact of prompt specificity on llm code generation. In *Proceedings of the 14th International Joint Conference on Natural Language Processing (AACL-IJCNLP)*, pages 2380–2402, 2025.

[12] Zane Fink^{+}, Konstantinos Parasyris, Praneet Rathi, Giorgis Georgakoudis, **Harshitha Menon**, and Peer-Timo Bremer. Hpac-ml: A programming model for embedding ml surrogates in scientific applications. In *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–16. IEEE, 2024.

[13] Jan Hückelheim, **Harshitha Menon**, William Moses, Bruce Christianson, Paul Hovland, and Laurent Hascoët. A taxonomy of automatic differentiation pitfalls. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 14(6):e1555, 2024.

[14] **Harshitha Menon**, Daniel Nichols^{+}, Abhinav Bhatele, and Todd Gamblin. Learning to predict and improve build successes in package ecosystems. In *Proceedings of the 21st International Conference on Mining Software Repositories (MSR)*, pages 531–542, 2024.

[15] Daniel Nichols^{+}, Aniruddha Marathe, **Harshitha Menon**, Todd Gamblin, and Abhinav Bhatele. Hpc-coder: Modeling parallel programs using large language models. In *ISC High Performance 2024*, pages 1–12, 2024.

[16] Daniel Nichols^{+}, **Harshitha Menon**, Todd Gamblin, and Abhinav Bhatele. A probabilistic approach to selecting build configurations in package managers. In *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–13. IEEE, 2024.

[17] Zane Fink^{+}, Konstantinos Parasyris, Giorgis Georgakoudis, and **Harshitha Menon**. Hpac-offload: Accelerating hpc applications with portable approximate computing on the gpu. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, pages 1–14, 2023.

[18] **Harshitha Menon**, James Diffenderfer*, Giorgis Georgakoudis, Ignacio Laguna, Michael O Lam, Daniel Osei-Kuffuor, Konstantinos* Parasyris, and Jackson Vanover^{+}. Approximate high-performance computing: A fast and energy-efficient computing paradigm in the post-moore era. *IT Professional*, 25(2):7–15, 2023.

[19] Garima Singh^{+}, Baidyanath Kundu^{+}, **Harshitha Menon**, Alexander Penev, David J Lange, and Vassil Vassilev. Fast and automatic floating point error analysis with chef-fp. In *2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 1018–1028. IEEE, 2023.

[20] James Diffenderfer*, Daniel Osei-Kuffuor, and **Harshitha Menon**. A framework for error-bounded approximate computing, with an application to dot products. *SIAM Journal on Scientific Computing*, 44(3):A1290–A1314, 2022.

[21] Zhimin Li^{+}, **Harshitha Menon**, Kathryn Mohror, Shusen Liu, Luanzheng Guo, Peer-Timo Bremer, and Valerio Pascucci. A visual comparison of silent error propagation. *IEEE Transactions on Visualization and Computer Graphics*, 30(7):3268–3282, 2022.

[22] **Harshitha Menon**, Konstantinos Parasyris*, Tom Scogland, and Todd Gamblin. Searching for high-fidelity builds using active learning. In *Proceedings of the 19th International Conference on Mining Software Repositories (MSR)*, pages 179–190, 2022.

[23] Konstantinos Parasyris*, James Diffenderfer*, **Harshitha Menon**, Ignacio Laguna, Jackson Vanover^{+}, Ryan Vogt, and Daniel Osei-Kuffuor. Approximate computing through the lens of uncertainty quantification. In *SC22: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–14. IEEE, 2022.

[24] Zhimin Li^{+}, **Harshitha Menon**, Kathryn Mohror, Peer-Timo Bremer, Yarden Livant, and Valerio Pascucci. Understanding a program's resiliency through error propagation. In *Proceedings of the*

26th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP), pages 362–373, 2021.

BRepro. [25] Konstantinos Parasyris*, Giorgis Georgakoudis, **Harshitha Menon**, James Diffenderfer*, Ignacio Laguna, Daniel Osei-Kuffuor, and Markus Schordan. Hpac: evaluating approximate computing techniques on hpc openmp applications. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, pages 1–14, 2021. **Best Reproducibility Award**.

[26] Zhimin Li+, **Harshitha Menon**, Dan Maljovec, Yarden Livnat, Shusen Liu, Kathryn Mohror, Peer-Timo Bremer, and Valerio Pascucci. Spotsdc: Revealing the silent data corruption propagation in high-performance computing systems. *IEEE Transactions on Visualization and Computer Graphics*, 27(10):3938–3952, 2020.

[27] **Harshitha Menon**, Abhinav Bhatele, and Todd Gamblin. Auto-tuning parameter choices in hpc applications using bayesian optimization. In *2020 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 831–840. IEEE, 2020.

[28] Konstantinos Parasyris*, Ignacio Laguna, **Harshitha Menon**, Markus Schordan, Daniel Osei-Kuffuor, Giorgis Georgakoudis, Michael O Lam, and Tristan Vanderbruggen. Hpc-mixpbench: An hpc benchmark suite for mixed-precision analysis. In *2020 IEEE International Symposium on Workload Characterization (IISWC)*, pages 25–36. IEEE, 2020.

[29] Seonmyeong Bak, **Harshitha Menon**, Sam White, Matthias Diener, and Laxmikant Kale. Multi-level load balancing with an integrated runtime approach. In *2018 18th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID)*, pages 31–40. IEEE, 2018.

[30] **Harshitha Menon**, Michael O Lam, Daniel Osei-Kuffuor, Markus Schordan, Scott Lloyd, Kathryn Mohror, and Jeffrey Hittinger. Adapt: Algorithmic differentiation applied to floating-point precision tuning. In *SC18: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 614–626. IEEE, 2018.

[31] **Harshitha Menon** and Kathryn Mohror. Discvar: Discovering critical variables using algorithmic differentiation for transient faults. In *Proceedings of the 23rd ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*, pages 195–206. ACM, 2018.

[32] Bilge Acun, Akhil Langer, Esteban Meneses, **Harshitha Menon**, Osman Sarood, Ehsan Totoni, and Laxmikant V Kalé. Power, reliability, and performance: One system to rule them all. *IEEE Computer*, 49(10):30–37, 2016.

[33] A Bastidas Fry, F Governato, A Pontzen, T Quinn, M Tremmel, L Anderson, H Menon, AM Brooks, and J Wadsley. All about baryons: revisiting sidm predictions at small halo masses. *Monthly Notices of the Royal Astronomical Society*, 452(2):1468–1479, 2015.

[34] **Harshitha Menon**, Lukasz Wesolowski, Gengbin Zheng, Pritish Jetley, Laxmikant Kale, Thomas Quinn, and Fabio Governato. Adaptive techniques for clustered n-body cosmological simulations. *Computational Astrophysics and Cosmology*, 2(1):1, 2015.

[35] Bilge Acun, Abhishek Gupta, Nikhil Jain, Akhil Langer, **Harshitha Menon**, Eric Mikida, Xiang Ni, Michael Robson, Yanhua Sun, Ehsan Totoni, et al. Parallel programming with migratable objects: Charm++ in practice. In *SC'14: Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 647–658. IEEE, 2014.

BPaper [36] Jonathan Lifflander, Esteban Meneses, **Harshitha Menon**, Phil Miller, Sriram Krishnamoorthy, and Laxmikant V Kalé. Scalable replay with partial-order dependencies for message-logging fault tolerance. In *2014 IEEE International Conference on Cluster Computing (CLUSTER)*, pages 19–28. IEEE, 2014. **Best Paper Award**.

[37] **Harshitha Menon**, Bilge Acun, Simon Garcia De Gonzalo, Osman Sarood, and Laxmikant Kalé. Thermal aware automated load balancing for hpc applications. In *2013 IEEE International Conference on Cluster Computing (CLUSTER)*, pages 1–8. IEEE, 2013.

BPaper Fin. [38] **Harshitha Menon** and Laxmikant Kalé. A distributed dynamic load balancer for iterative applications. In *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, pages 1–11, 2013. **Best Student Paper Award Finalist**.

[39] **Harshitha Menon**, Nikhil Jain, Gengbin Zheng, and Laxmikant Kalé. Automated load balancing invocation based on application characteristics. In *2012 IEEE International Conference on Cluster Computing*, pages 373–381. IEEE, 2012.

Workshop Papers & Technical Reports

[40] Gregory Bolet, Giorgis Georgakoudis, **Harshitha Menon**, Konstantinos Parasyris, Niranjan Hasabnis, Hayden Estes, Kirk Cameron, and Gal Oren. Can large language models predict parallel code performance? In *Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing (HPDC)*, pages 1–6, 2025.

[41] Gautam Singh*, Arjun Guha, Bhavya Kailkhura, and **Harshitha Menon**. Can test-time compute help llms write low-resource parallel code better? In *NeurIPS 2025 Fourth Workshop on Deep Learning for Code*, 2025.

[42] Keita Teranishi, **Harshitha Menon**, William F Godoy, Prasanna Balaprakash, David Bau, Tal Ben-Nun, Abhinav Bhatele, Franz Franchetti, Michael Franusich, Todd Gamblin, et al. Leveraging ai for productive and trustworthy hpc software: Challenges and research directions. In *LLM4HPC Workshop at International Conference on High Performance Computing (ISC)*, pages 615–625. Springer, 2025.

[43] Todd Gamblin, **Harshitha Menon**, Tom Scogland, and Mathew Legendre. Build: Binary understanding and integration logic for dependencies final report. Technical report, Lawrence Livermore National Laboratory (LLNL), Livermore, CA (United States), 2023.

[44] Jan Hueckelheim, **Harshitha Menon**, William S Moses, Bruce Christianson, Paul Hovland, and Laurent Hascoet. A short review of automatic differentiation pitfalls in scientific computing. In *ICML 2023 Workshop on Differentiable Almost Everything*, 2023.

[45] JA Hittinger, PG Lindstrom, H Bhatia, PT Bremer, DM Copeland, KK Chand, AL Fox, GS Lloyd, H Menon, GD Morrison, et al. Variable precision computing. Technical report, Lawrence Livermore National Lab.(LLNL), Livermore, CA (United States), 2019.

[46] Michael O Lam, Tristan Vanderbruggen, **Harshitha Menon**, and Markus Schordan. Tool integration for source-level mixed precision. In *2019 IEEE/ACM 3rd International Workshop on Software Correctness for HPC Applications (Correctness)*, pages 27–35. IEEE, 2019.

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